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DEPARTMENT OF MECHATRONICS, ROBOTICS AND DIGITAL MANUFACTURING

MECHATRONIC SYSTEMS 1
HOMEWORK REPORT

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1. Introduction

Magnetic levitation (maglev) systems allow objects to be suspended without mechanical support by using magnetic forces. The levitating disk is influenced by electromagnetic forces, gravity, and damping. The system is inherently unstable and requires precise control, typically implemented using a PID controller, to maintain the desired levitation height.

In this report, a MATLAB Simulink model is used to simulate the levitation dynamics. The disk mass is 5 kg, and the effects of varying damping and PID parameters on the system's motion are analyzed.

2. Aim

The aim of this work is to study the behavior of a magnetic levitation system with a levitating disk of mass 50 kg. Specifically, this study focuses on:

1. Investigating the effect of damping on the system's stability and oscillatory behavior.
2. Analyzing the effect of PID (Proportional-Integral-Derivative) controller parameters on the system to achieve stable levitation without bouncing or overshoot.

3. System Modeling

The magnetic levitation system can be represented by the following dynamics:

$$m\ddot{z} = F_m(z, i) - mg - b\dot{z}$$

Where:

- $m = 50$ kg (mass of the disk)
- z = displacement of the disk
- F_m = magnetic force as a function of position z and coil current i
- g = gravitational acceleration (9.81 m/s^2)
- b = damping coefficient

- $\dot{z}, \ddot{z}$ = velocity and acceleration of the disk

A PID controller is used to regulate the coil current i to maintain the following disk at the desired position z_{ref} .

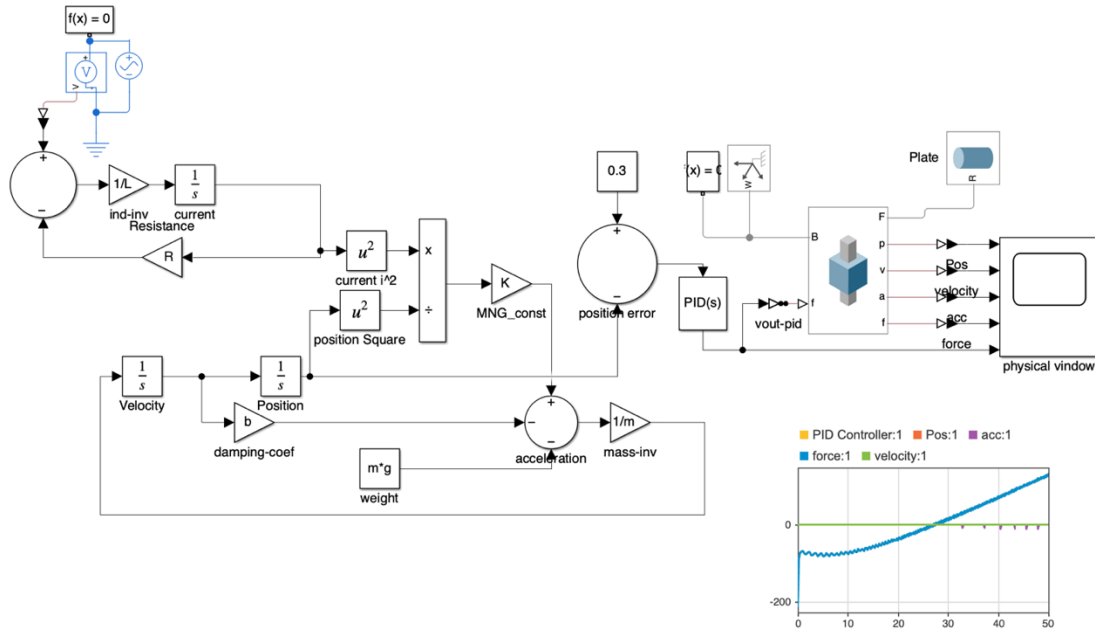


Figure 1 Maglev Model

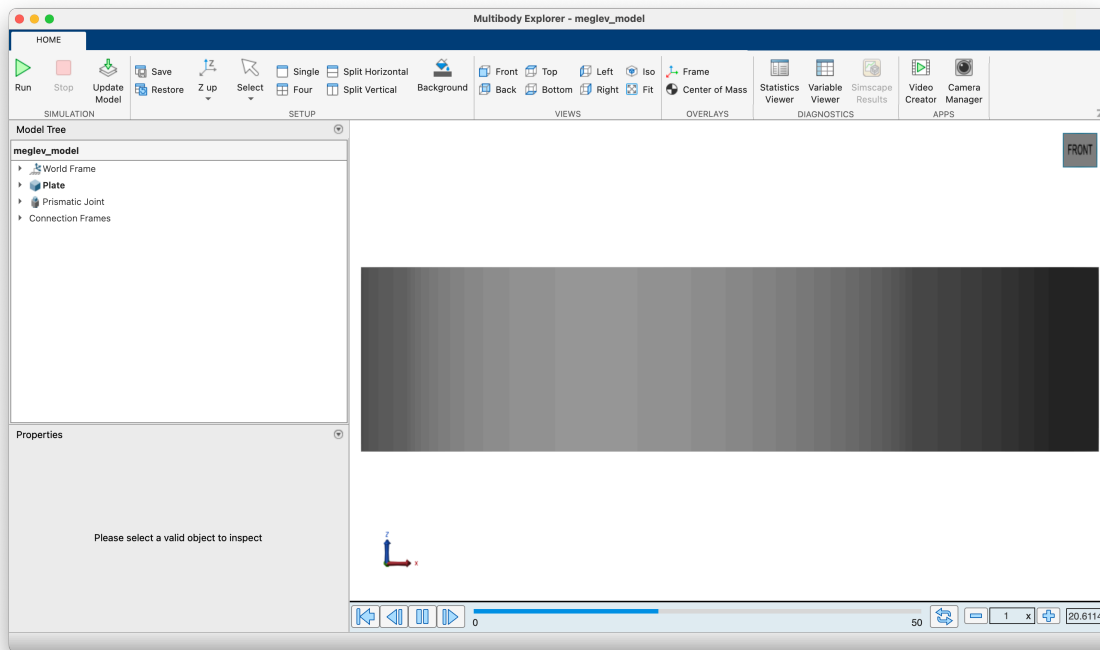


Figure 2 image of the plate being simulated.

4. Tasks

The report evaluates the following tasks derived from the aim:

Effect of Damping:

- Study how different damping coefficients affect the levitation stability and oscillation of the disk.
- Identify the optimal damping value to reduce bouncing without slowing system response excessively.

Effect of PID Parameters:

- Analyze the system response for varying PID parameters (Proportional K_p , Integral K_i , Derivative K_d).
- Achieve stable levitation without bouncing or overshoot.
- Determine the PID settings that give the best balance between fast response and stability.

5. Simulation Initial Conditions (Base Line)

- Disk initial position: 0.00 m
- Damping: 0.1
- K_p :100
- K_i :20
- K_d :20
- Initial velocity: 0 m/s
- Mass: 50 kg

6. Results and Analysis

6.1 Effect of Damping

In the model the Damping seems to have minimal effect on the result below 2 cases are compared.

Case 1 Damping is 5

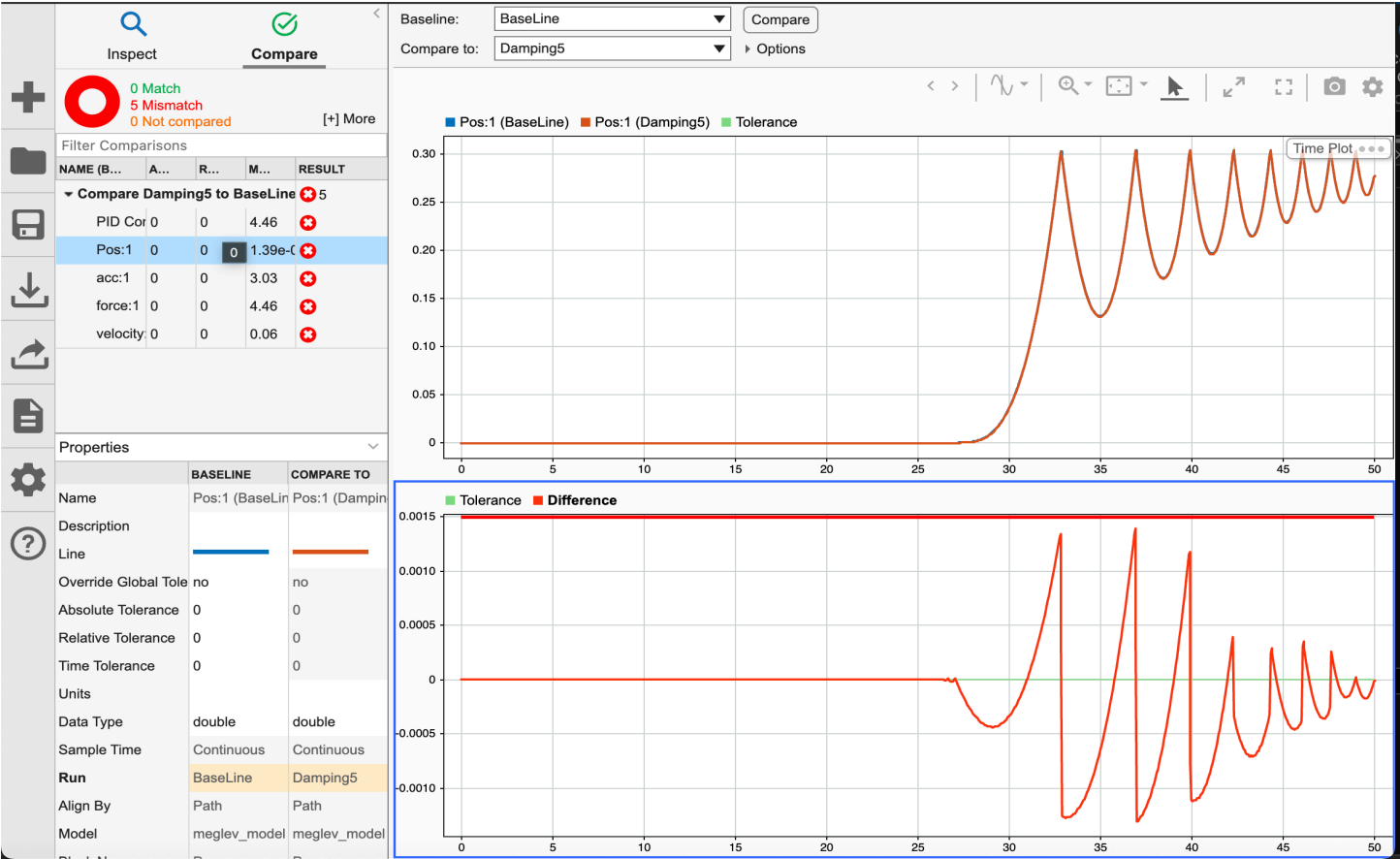


Figure 3 Comparison of Position when damping is 5

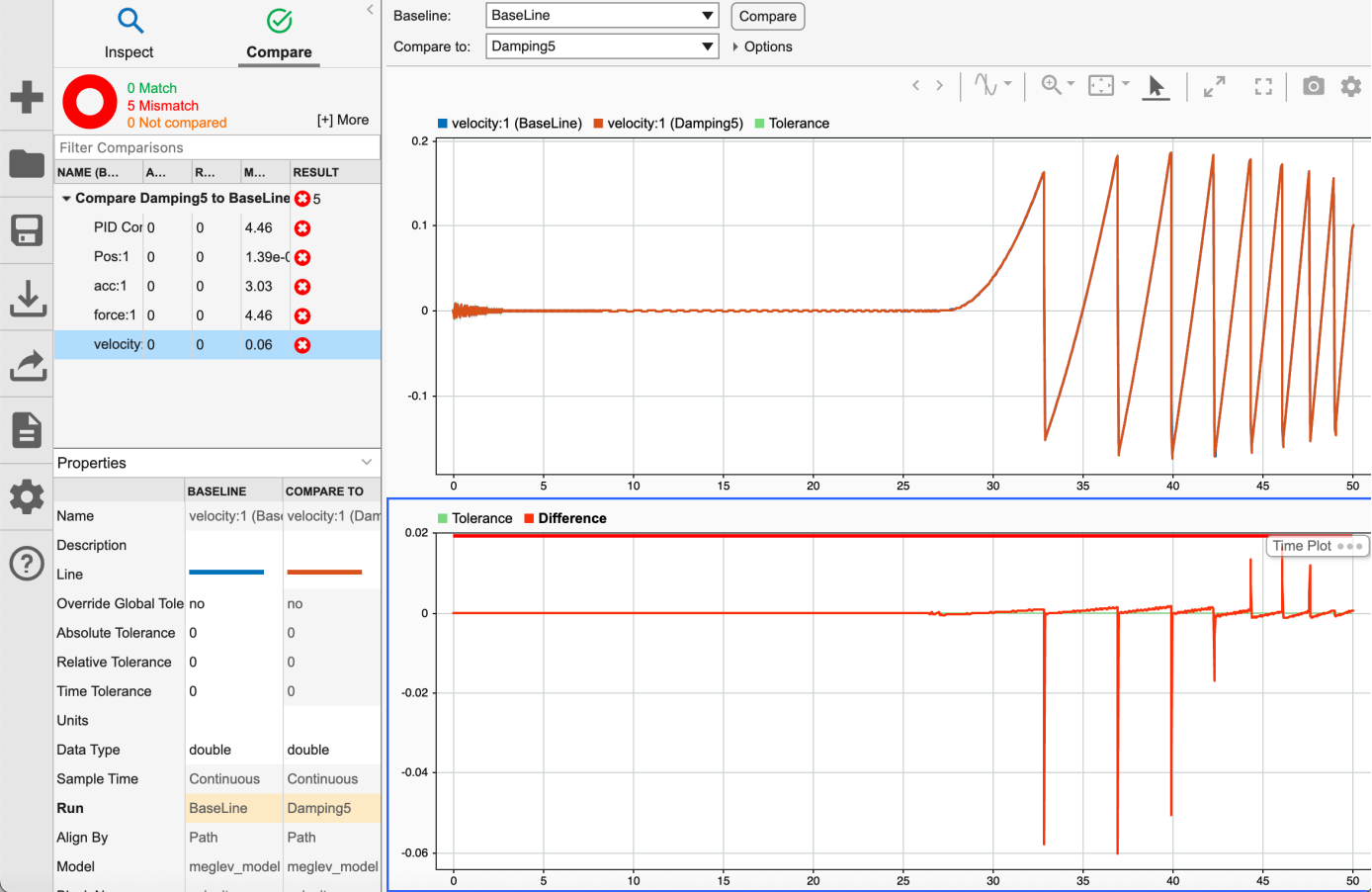


Figure 4 Comparison of Velocity Damping is 5

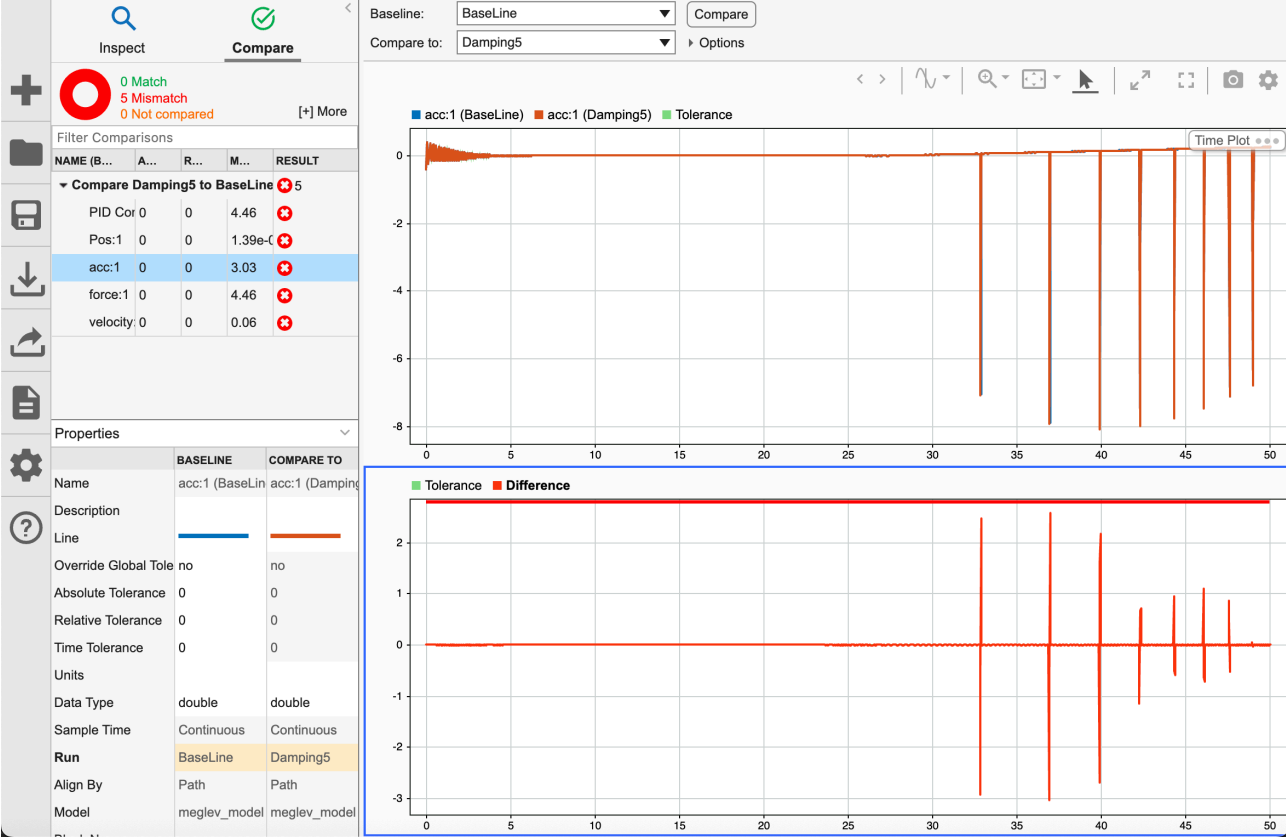


Figure 5 Comparison of Acceleration when damping is 5

Case2 Damping is 50

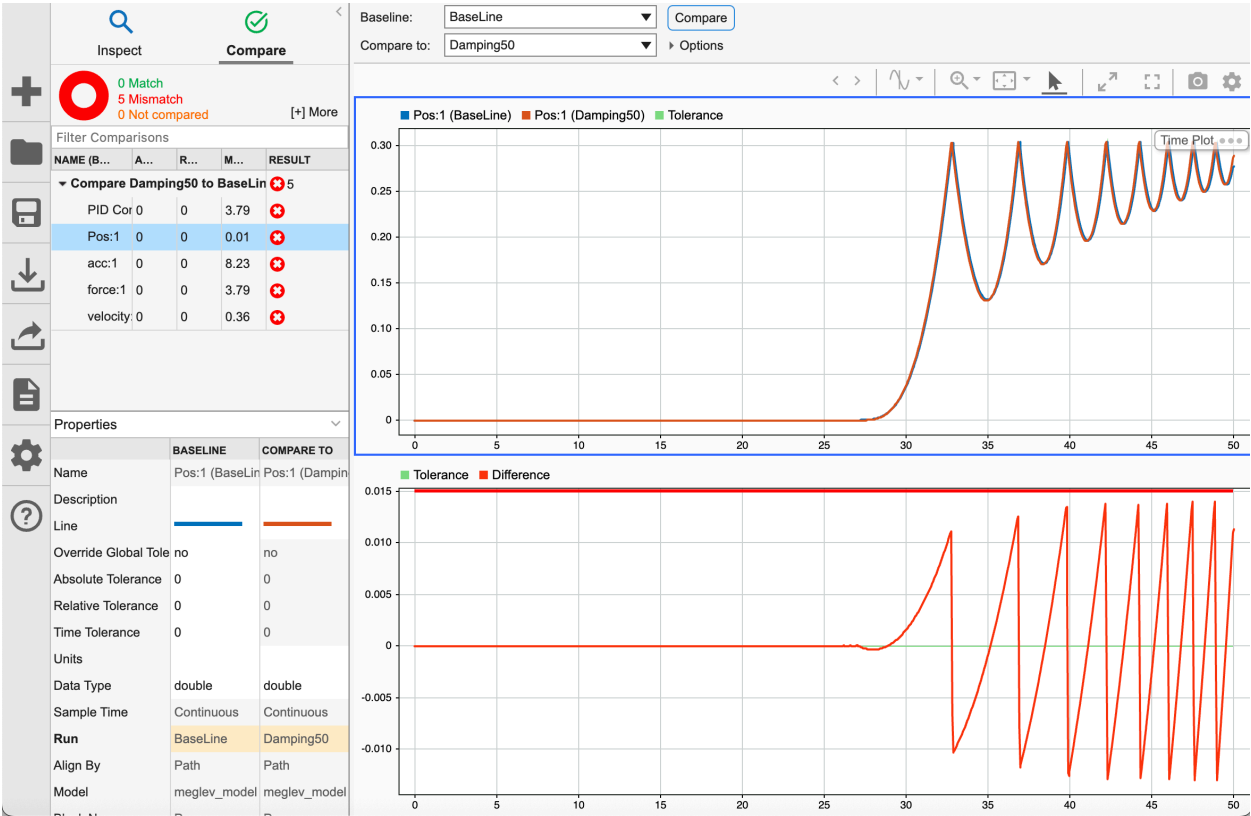


Figure 6 Comparison of Position Damping is 50

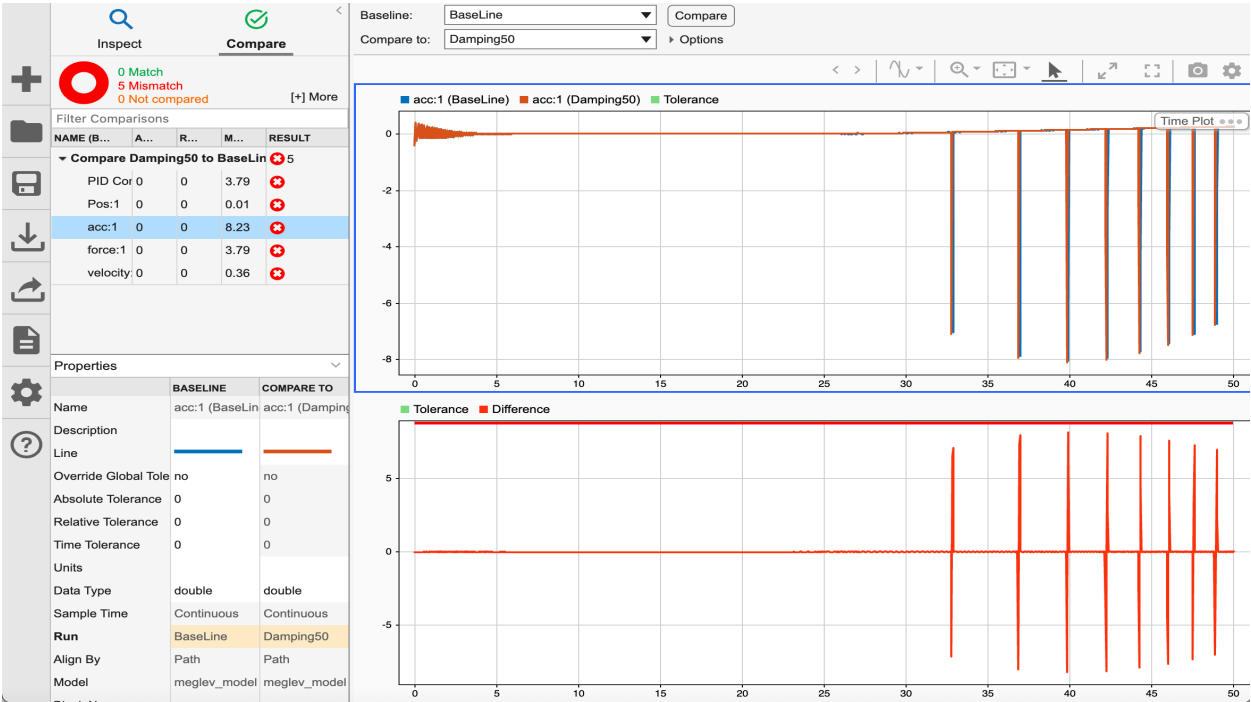


Figure 7 Comparison of Acceleration when Damping is 50

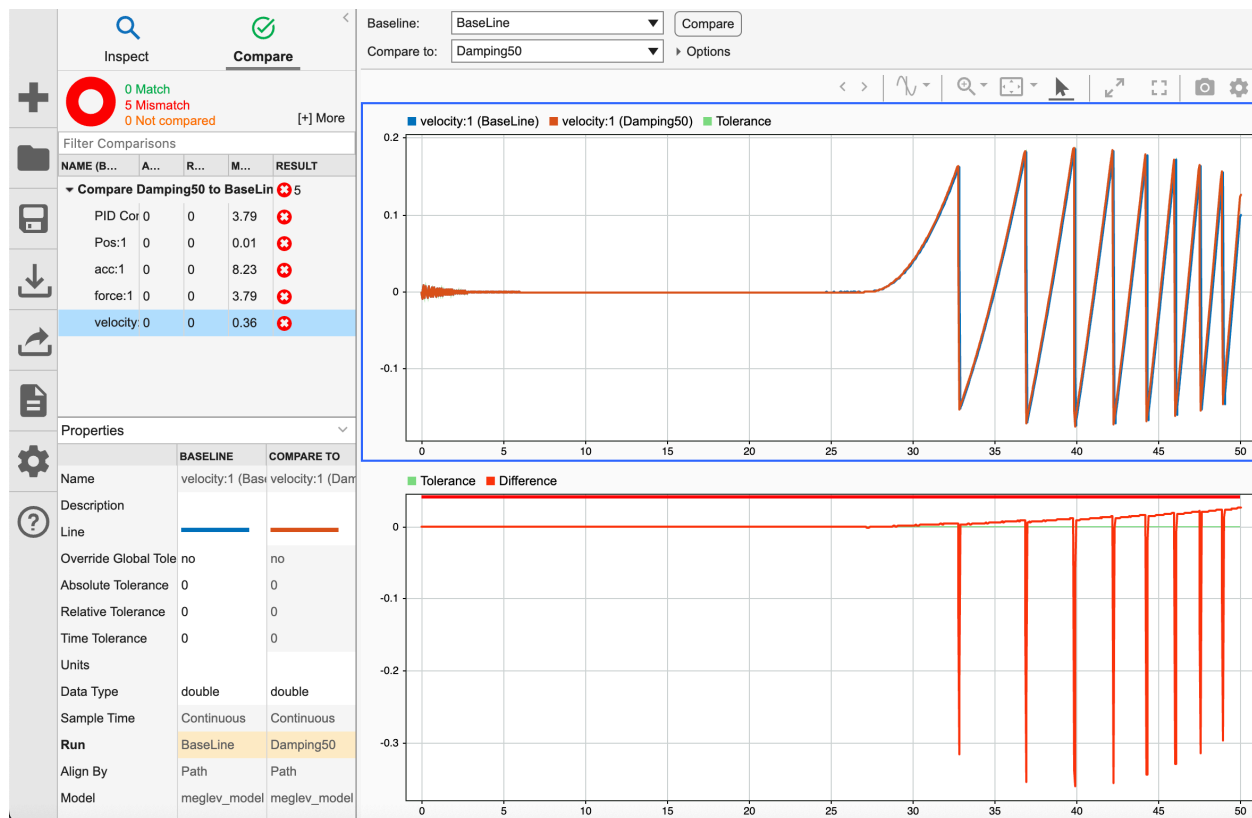


Figure 8 Comparison of velocity when damping is 50

Observation: Damping seems to have minimal effect on the final position of the levitating plate.

6.2 Effect of PID Parameters

- Increasing K_p → reduces steady-state error but may induce oscillations.
- Increasing K_i → eliminates steady-state error, may increase overshoot if too high.
- Increasing K_d → reduces oscillations, improves stability.

Case1 PID Tune setting Baseline, PID(800,80,80), and PID(3000,250,250)

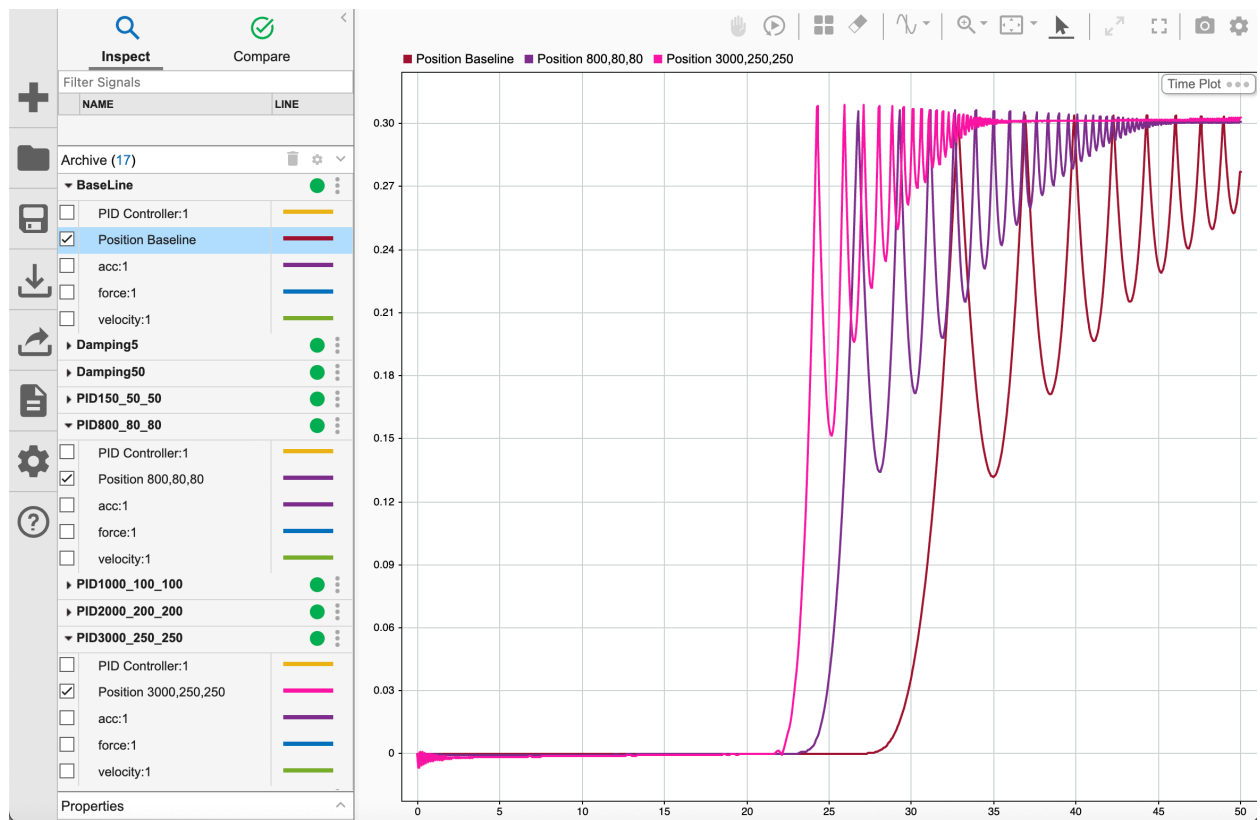


Figure 9 Comparison of Position with different Tune setting

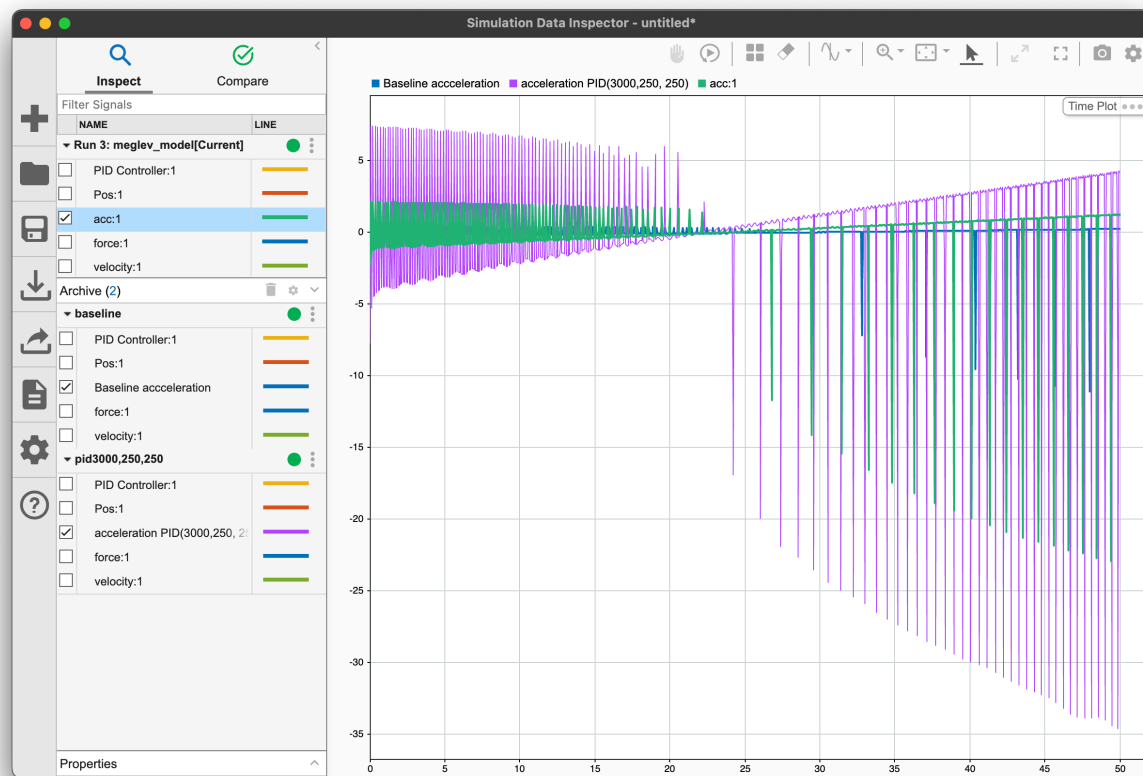


Figure 10 Comparison of Acceleration with different Tune setting

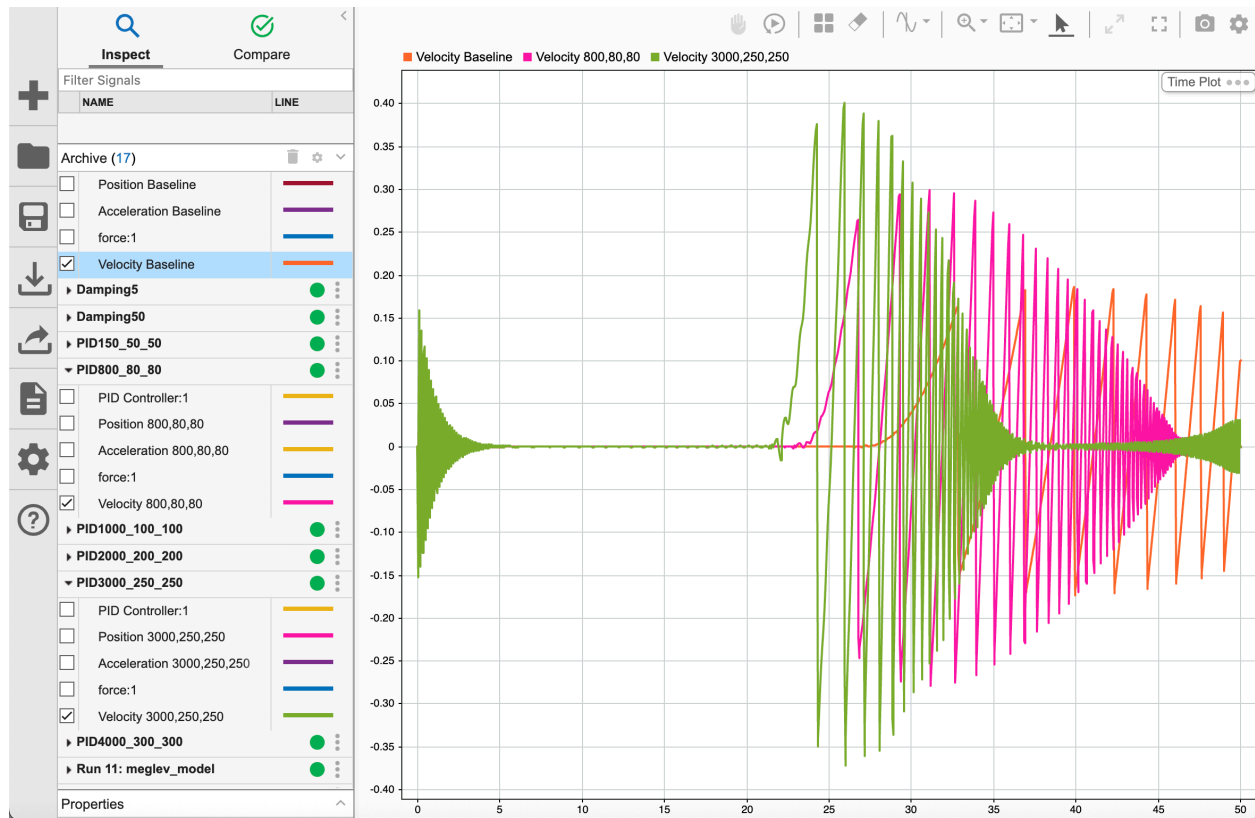


Figure 11 Comparison of Acceleration with different Tune setting

Observation: the PID tune settings affects the system properties and the comparison curves show that the as the tune settings are increased the stability in the system is achieved faster.

7. Conclusion

The MATLAB Simulink simulation of the magnetic levitation system demonstrates that:

1. **Damping** has no crucial effect on the system.
2. **PID parameters** critically influence system stability. Proper tuning achieves stable levitation with minimal overshoot and bouncing.

This study highlights the importance of control system design in practical magnetic levitation applications.